

GOWDUL AALAM

Senior Agentic AI Engineer · LangGraph · LangChain · LangSmith · Multi-Agent Systems

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PROFESSIONAL PROFILE

Senior Agentic AI Engineer with 4+ years building production-grade LLM-powered systems for enterprise clients (Verizon, Wells Fargo). Deep hands-on expertise with LangGraph, LangChain, and LangSmith — designing stateful, multi-step agent workflows with structured state, memory strategies, and controlled tool-calling execution loops. Proven track record taking complex agentic systems from POC to production: *Coder Buddy* (LangGraph Planner→Architect→Coder pipeline with full LangSmith observability) and *Automated Event Planner* (CrewAI autonomous multi-agent system with human-in-the-loop gates). Writes clean, modular, production-grade Python; strong async and backend system design skills. Fully remote, immediately available.

CORE TECHNICAL SKILLS

Agentic Frameworks: LangGraph · LangChain · CrewAI · Multi-Agent Orchestration · Tool/Function Calling · Human-in-the-Loop · MCP

Observability & Eval: LangSmith (tracing, debugging, evaluation, monitoring) · Token usage tracking · Latency profiling · Cost optimisation

LLM & RAG: OpenAI API · Anthropic API · Groq · Mistral · Ollama · Hugging Face · RAG Pipelines · FAISS · Chroma · Embeddings

State & Memory: Structured state management · Context propagation · Persistence strategies · Pydantic structured outputs · Async Python

Backend & APIs: Python (FastAPI) · REST APIs · Microservices · Java (Spring Boot) · SQL · Docker · Webhook handling

Cloud & Data: AWS (EC2, S3, Lambda, Kinesis Firehose) · DynamoDB · PostgreSQL · Neo4j · ELK Stack

Dev Tools: Git / GitHub · GitLab · Docker · CI/CD · Prompt Engineering · uv

PROFESSIONAL EXPERIENCE

Senior Analyst — Generative AI & Agentic Systems · AccentureDec 2024 – Present · Coimbatore, India (Remote)

Enterprise AI for Verizon & Wells Fargo — LLM Pipelines · Agentic Workflows · Observability

- ▶ Designed and implemented stateful, multi-step LLM agent workflows processing live JSON feeds (Verizon, Wells Fargo) — automating root-cause detection, pattern mining, escalation identification, and callback intent extraction across thousands of call-centre records daily; each pipeline step uses structured tool-calling with controlled execution flow.
- ▶ Built structured state and context management across agent lifecycles — integrated DynamoDB/RDS output layers and analytics dashboards, reducing issue-resolution speed by 40% and root-cause discovery time by 40%.
- ▶ Engineered LLM evaluation and benchmarking framework across OpenAI, Anthropic, and open-source models — measuring accuracy, coherence, latency, and cost/token; selected optimal model reducing processing time by 35%. Mirrors LangSmith-style evaluation disciplines.
- ▶ Implemented LLM reliability mechanisms: structured output guards, rate-limiting layer (100 req/day cap), and observability hooks — reducing server load by 40% and enabling full cost tracking across AI workloads.
- ▶ Prototyped AI-powered automated API testing agent (Karate Labs) with multi-step API chaining and LLM-generated test cases — POC to working system in under 3 weeks; cut manual QA effort by ~40%.
- ▶ Owned full model pipeline, LLM API layer, and containerisation boundary end-to-end; collaborated with DevOps/Platform team on cloud provisioning and CI/CD.

SDE II — Backend & Integration Engineer · Mindgate SolutionsApr 2024 – Nov 2024 · Chennai, India (Remote)

- ▶ Built fault-tolerant API automation workflows integrating Middle Eastern banking APIs (Efawateer, Sadad) — reduced transaction processing errors and improved throughput by ~20%.

- ▶ Automated end-to-end transaction failure detection and resolution workflows via API integrations with core banking systems.

Software Developer — AI & Backend Engineering · Senseforth.ai Solutions Jun 2021 – Jan 2024 · Bangalore, India

- ▶ Architected AWS Kinesis Firehose → S3 → ELK Stack streaming pipeline — eliminated manual upload processes, improved data ingestion latency by ~50%; promoted from POC to production.
- ▶ Built LLM-powered Text-to-Speech automation integrated into chatbot application using Python and Java — boosted user engagement by 30%.
- ▶ Delivered Salesforce–WhatsApp workflow automation (Twilio APIs, POC to production): automated lead notifications, messaging workflows, and two-way interactions via webhook handling.

KEY AGENTIC AI PROJECTS

Coder Buddy — AI Multi-Agent Developer Productivity Tool — *LangGraph · FastAPI · Groq API · LangSmith · Python*

- ▶ Built production-grade multi-agent coding assistant using **LangGraph: Planner → Architect → Coder stateful pipeline** — each agent node manages structured state, passes context across steps, and calls specialised tools to translate natural-language requirements into complete, file-by-file project scaffolds end-to-end.
- ▶ Integrated **LangSmith for full agent observability**: token usage tracking, latency profiling, and chain execution tracing across all agent steps — enabling debugging, evaluation, and reliability monitoring identical to the JD requirements.
- ▶ Used Groq API for open-source LLM backends with reproducible deployments via uv — demonstrated LLM cost and latency optimisation in a live multi-agent system; reduced prompt-to-working-code time by ~60%.

Automated Event Planner — Autonomous Multi-Agent Workflow — *CrewAI · Python · Pydantic · SerperDev · ScrapeWebsite Tools*

- ▶ Designed fully autonomous CrewAI multi-agent system (Venue Coordinator, Logistics Manager, Marketing & Communications) — **three specialised agents with distinct memory and tool configurations** executing real-world workflows end-to-end.
- ▶ Implemented async task execution for parallel agent workflows, Pydantic-structured JSON outputs with validation guards, and **human-in-the-loop gates** on critical decision points — demonstrating production-grade reliability and structured output patterns.
- ▶ Configured per-agent web search and scraping tools with controlled execution loops; system autonomously retrieves, evaluates, and acts on live external data with no manual intervention.

EDUCATION

B.E. Computer Science Engineering · Karpagam College of Engineering CGPA: 8.75 / 10.0 · May 2022

AVAILABILITY & WORK AUTHORISATION

Availability: Immediate · Fully Remote · EU time-zone overlap (CET/CEST compatible) **Work Auth:** Full-time or B2B contract · Based in India — no EU work authorisation required for remote engagement